Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AUOE02]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

AUTOMOTIVE SAFETY

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain briefly about various design techniques adopted while designing the car body with necessary sketches.	**07**
	b.	Discuss briefly about safety sandwich construction and properties of sandwich panels with diagram.	**07**
2.	**a.**	Explain briefly in detail about safety concepts with flow charts.	**07**
	b.	Illustrate briefly about the deformation behavior of vehicle body with neat sketches.	**07**
3.	**a.**	Define seat belt and its function .Explain its types with seat belt tightners sketches.	**07**
	b.	Give explanation about the function and operation of air bag with neat diagrams.	**07**
		OR	
4.	**a.**	Explain briefly about Bumper Design for safety with diagrams.	**06**
	b.	Give the detailed explanation of collapsible steering column with sketches.	**08**
5.	**a.**	Explain different collision warning system with flow charts and diagrams.	**08**
	b.	Give the detailed clarity about layout of object detection technology.	**06**
		OR	
6.	**a.**	Explain about edge detection algorithm with neat sketch.	**06**
	b.	Explain Automatic braking control system with circuit diagram.	**08**
7.	**a.**	Explain the following	
		i. Steering and mirror adjustment	**03**
		ii. Central locking system with neat sketch	**04**
	b.	Discuss briefly about Tire Pressure control system with sketches.	**07**
		OR	
8.	**a.**	Explain clearly about the Rain sensors with diagram.	**07**
	b.	Explain garage door opening system with neat diagram.	**07**